Turbolift car Specification

Document number	SHV-TR-SPEC-0101
Revision	C
Status	APPROVED
Classification	INTERNAL
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Date	2022-04-12

1. Purpose

This specification defines the performance, interface and verification requirements for the turbolift car (TR-TLC) of the Internal Transit on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-TR-02 (WBS 1.15.2), owned by Vela Wade.

The prime supplier for flight hardware is Anaxes Turbolift Company. Responsible engineer: Esme Hesk.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-TR-DWG-0201	Turbolift car general arrangement drawing
SHV-TR-BOM-0301	Turbolift car bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-PD-TR	Interface Control Document: Power Distribution / Internal Transit
SHV-ICD-CN-TR	Interface Control Document: Computer Core & Data Network / Internal Transit
SHV-ICD-SF-TR	Interface Control Document: Superstructure & Frame / Internal Transit

4. Description

Internal transit moves people and goods around the station: turbolift cars in vertical shafts and a tram that runs around every latitude band.

The turbolift car is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
TR-TLC-R01	The turbolift car shall provide a car capacity of 39 persons.	39 persons	±0.5%	Analysis
TR-TLC-R02	The turbolift car shall provide a car speed of 11.7 m/s.	11.7 m/s	±2.5%	Demonstration

5.1 General requirements

ID	Requirement	Verification
TR-TLC-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
TR-TLC-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
TR-TLC-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
TR-TLC-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
TR-TLC-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the TR test-report series.

8. Revision history

Rev	Date	Description	Author
A	2021-06-26	Initial release.	Esme Hesk
B	2021-11-17	Editorial corrections.	Esme Hesk
C	2022-04-12	Incorporated interface changes.	Esme Hesk